Enrollment No._____

Ganpat University

Acharya Motibhai Patel Institute of Computer Studies

Dual(BCA-MCA)

First Semester

Subject: U51C4IDE - Introduction to Digital Electronics

Internal Audit Examination

Date : 16/10/2026 Time: 9:15 AM to 10:15 AM Total: 20 Marks (Theory)

Assessment Pattern

Bloom's Taxonomy Based Assessment Pattern :

BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create;

Bloom's Category	Section	Description	Marks	Total Marks
BTL1, BTL2	05 question, 1 mark each (Short questions / MCQ/Fill in the blank).	Choose/ Fill ups/ One word, 1-2 Sentence.	05	20
BTL3, BTL4, BTL5, BTL6	03 out of 05 questions, 05 mark each	150-200 Words	15	

Question Paper

Section – A				
Sr.No.	**Answer the following**	**Marks**	**CO**	**Bloom's Taxonomy Pattern**
1	Define Logic Circuit.	01	CO1	BTL1
2	Which type of memory is volatile? (a) ROM (b) EPROM (c) EEPROM (d) RAM	01	CO3	BTL1
3	A NOT gate is also called an _____ gate. (a) OR (b) Inverter (c) AND (d) Buffer	01	CO1	BTL2
4	How many output lines does a 3-to-8 decoder have? (a) 3 (b) 6 (c) 8 (d) 16	01	CO2	BTL2
5	Which register operation shifts the stored bits toward the most significant bit position? (a) Right shift (b) Left shift (c) Parallel loading (d) Reset operation	01	CO4	BTL3
Section – B **Answer any 03 from the following**				
6	What is Logic Gate? Explain all basic gates.	05	CO1	BTL2
7	Explain the working of an edge-triggered D flip-flop with a neat logic diagram and truth table.	05	CO4	BTL3
8	Explain Universal Gates with a suitable diagram and truth table.	05	CO1	BTL3
9	Differentiate between RAM and ROM. Explain SRAM and DRAM with their characteristics and applications.	05	CO3	BTL4
10	What is a Multiplexer? Explain its working with truth table and diagram.	05	CO2	BTL3

**** End of the Paper ****